

Session 22: Discover & Define

You've spent 21 sessions learning **HOW AI works**. Now you'll put it all together and produce something you can show, share, and keep. Over the next 4 sessions, **you will write a real research paper** investigating how AI is being used in a domain you pick - the same kind of paper real researchers write.

This Session: you pick your topic, sharpen your question, and find your sources.



Objectives:

By the end of this Session, you'll have done **all** of these:

01

Pick YOUR Domain

Choose what part of AI you'll investigate

02

Sharpen Your Research Question

Using the Three-Part Test

03

Understand the Paper Structure

The 8 sections you'll write

04

Start Source Discovery

Collect 8–12 candidates with Perplexity

05

Commit to Your Hands-On Plan

What you'll explore next session

06

Get Clear on Homework

Abstract, intro, source filtering, and Related Work

The Capstone Topic

Your topic is fixed at the big level:  **Applications of AI in Various Domains**. But **YOU** pick the specific area. Choose what you care about:

Healthcare

Allergies, dyslexia, hearing aids

Sports

Coaching apps, performance analysis

Music

Making songs, music tutors

Education

Homework helpers, math tutors

Wildlife

Identifying animals, conservation

Accessibility

Helping people with disabilities

 **One rule:** Pick a domain that interests you AND that you're comfortable researching. Avoid topics that feel too heavy or upsetting. If you're not sure, talk it through with your mentor.

What This Paper IS and ISN'T

✓ What It IS

- A real investigation into how AI is being used in your area
- Two threads - reading research + doing a hands-on test
- A synthesis - weaving the two together

✗ What It ISN'T

- Not a fancy engineering paper with statistics
- Not a claim that you built new AI tech
- Not just playing with a tool - your tool exploration is evidence for your paper, not the paper itself
- Not someone else's words rearranged - every sentence must be in **YOUR voice**

① You're investigating. Like a real journalist or researcher.



The Personalization Moment

Your mentor has been watching what you got excited about across 21 sessions. They're going to share 2–3 paper angles they think fit you. Listen to each one carefully before you choose.

Listen First

Hear all 2–3 angles before making any decision. Each one has been chosen with you in mind.

Ask Questions

If something isn't clear about an angle, ask your mentor to explain more before you commit.

Then Choose

Pick the angle that excites you most - you'll be spending 4 sessions on it!

How to Choose Your Angle

For each angle proposed, your mentor will tell you:



Domain & Focus

Exactly what we'd study



Why It Matches YOU

Based on what your mentor has observed across 21 sessions



The Research Question

The kind of question you'd answer



Existing Sources

What's already been written on this topic



Hands-On Plan

What you'd actually test next session

-  Pick the one you **actually want to investigate**, not the one that sounds the smartest. You're going to spend four sessions on this - pick what holds your interest.

What Makes a GOOD Research Question?

A topic is BROAD. A research question is SHARP.

A great research question passes **3 tests**:

SPECIFIC

Names a domain, a specific use, and what's being investigated

ORIGINAL

You're investigating, not just summarizing one article

ACHIEVABLE

You can answer it in 4 sessions



Sharpening a Research Question

Watch how a vague topic becomes a sharp question:

Too Broad ❌	Sharpened ✅
AI in Healthcare	<i>"How is AI helping families manage food allergies - and what concerns about bias and safety come with it?"</i>
AI in Education	<i>"How is AI being used to help kids practice math - and could it make kids depend on it too much?"</i>
AI in Music	<i>"How is AI being used to make music that matches different emotions - and what does that mean for music class?"</i>
AI in Sports	<i>"How is AI being used in basketball training apps for kids - and does it actually help them get better?"</i>

📌 See the pattern? Each one names: **domain** · **specific use** · **what to investigate**.

Mini Activity 1 - Sharpen YOUR Question!

Sharpen YOUR research question step by step:

Start with your domain

Write it down: _____

Add a specific use

Write it down: _____

Add what you'll investigate

Write it down: _____

Combine into ONE sharp question

Write it down: _____

Now test it with the 3 questions:

Specific?	Original?	Achievable?
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The Paper Structure - 8 Sections

Your paper will have **eight sections** plus an AI Attribution Statement and References.

1 - Abstract

One paragraph summary of everything

2 - Introduction

Why your topic matters

3 - Related Work

What other people have found

4 - Methodology

How YOU investigated

5 - Bias & Fairness

Who AI works for, who it fails

6 - Safety & Governance

Risks and rules

7 - Results

What you found

8 - Conclusion & Future Work

What it means + what's next

The 8 Sections Quick Tour

A peek at each section - when you write it and what it does:

Section	What It Does	When You Write It
Abstract	Summarizes everything	Placeholder today, final S24
Introduction	Hook + context + question	Outline today, prose in homework
Related Work	What others found	Homework after S22
Methodology	How you investigated	Session 23
Bias & Fairness	Who AI fails	Session 24
Safety & Governance	Risks + rules	Session 24
Results	Your findings	Started S23, finished S24
Conclusion	What it means	Session 24

Mini Activity 2 - Pick Your Tool 🛠️

Your paper needs a hands-on exploration in Session 23. Time to pick the AI tool you'll investigate with.

Tool	Best For
 Teachable Machine	Training your own image or sound classifier
 ChatGPT / Gemini	Testing AI conversations, prompts, generations
 ElevenLabs	Voice and audio explorations
 Lovable	Building a quick app to investigate
 NotebookLM	Comparing AI responses to source material
 Veo 3 / Pictory	Video and multimedia explorations

 My tool:

 What I'll test:

 What I'll watch for:

Finding Sources - The 3 Tool Workflow

You need to find **5–8 REAL sources**. Use this 3-tool workflow - each tool has a specific job. Use them in order.

Perplexity

Discover candidates - collect 8-12 sources with citations

Gemini

Compare and filter - narrow down to the most relevant

NotebookLM

Verify content - confirm each source is real and accessible

We'll do **Step 1 together**. Steps 2 and 3 are part of your homework so you have time to do them right.

Step 1 - Perplexity

What Is Perplexity?

A search tool that gives you answers AND shows you the real sources behind every answer. Every claim has a link → click it → see the actual source.

Today's goal with Perplexity:

Collect **8–12 candidate sources** about your research question.

Sample search: *"AI in food allergy management for children"*



Don't pick the first sources you see. Look for variety - academic, news, expert blogs.

The Source Credibility Checklist

For each candidate source, ask these five questions. Pass threshold: **4 of 5**. Drop sources that fail more than one check.

Question	If...
Is the author named?	Author has a name + their job/background
Is the publisher trustworthy?	Journal, university, government, big news, or expert site
Is it from the last 3 years?	AI changes fast - older = risky
Can you find the original?	Not just a summary of a summary
Will NotebookLM load it?	The link works and the content is real

Sources to AVOID

Some sources fail the credibility check almost always:

✗ Random Blogs

No named author - skip them

✗ Wikipedia

Use it for *orientation*, then cite what IT cites - never cite Wikipedia itself

✗ AI-Generated Articles

Without human verification - not acceptable

✗ Pay-to-Publish Journals

Not real peer review

✗ Social Media Posts

Not primary sources

✗ Company Advertising

Disguised as research - watch out

⚠ **Important reminder:** AI can **HALLUCINATE sources**. ALWAYS verify a source actually exists before adding it to your list.

Hands-On Activity - Run YOUR Source Discovery!

Time to actually start. With your mentor, work through **Step 1 only**:

Open Perplexity

search your research question

Collect 8–12 candidate sources

Save titles, authors, dates, and URLs

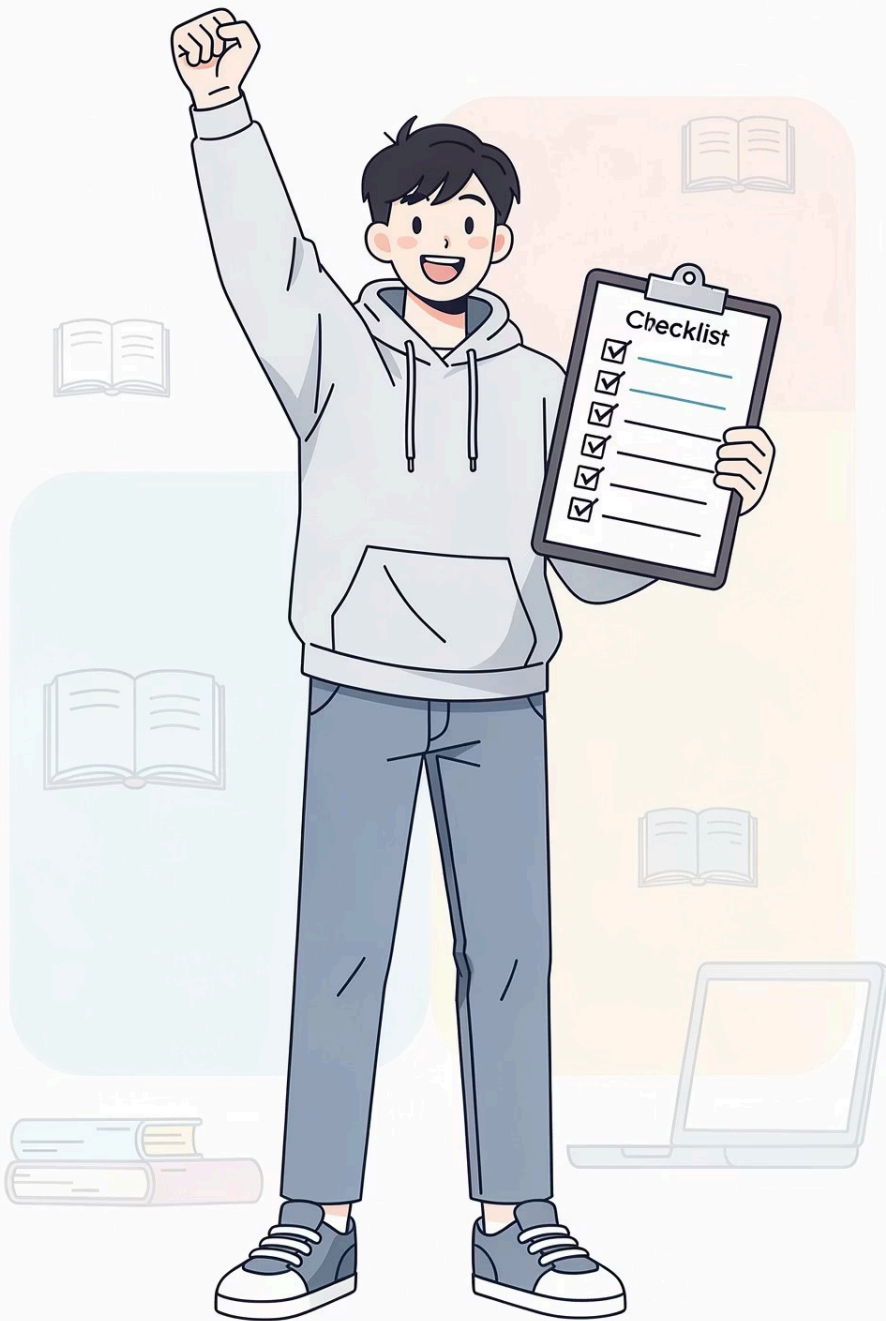
Apply the Credibility Checklist to each

Cut sources that fail 2+ tests

Save your shortlist in a document

You'll bring this to your homework

- ① What you should leave today's session with: A list of 8–12 candidates (or fewer if you've already cut some). You'll filter to 5–8 in homework using Gemini, then verify each one with NotebookLM.



What We Did Today

Domain Picked

Sharpened your research question using the 3-Part Test

Paper Structure

Understood all 8 sections of your research paper

Source Discovery

Collected 8–12 Perplexity candidates and applied the Credibility Checklist

Hands-On Plan

Committed to your tool and what you'll explore next session

You're officially writing a research paper.

Your Homework

Before Session 23, complete **ALL** of these:



Filter Your Sources with Gemini

Paste your 8–12 candidates into Gemini and ask which 5–8 are most useful for a research paper, and why each one matters. Combine its recommendations with your own judgment to land on your final list.

Verify Each One with NotebookLM

Add each finalist to NotebookLM and ask whether it actually discusses your topic. If it won't load or turns out to be off-topic, cut it and pull a replacement from your earlier candidates.



Write a One-Sentence Summary of Each Source

In your own words, capture what each source contributes to your question – not a quote, not the first line restated. If you can't say it without copying, you don't understand it yet.



Sketch Your Abstract – ~80 Words Placeholder

Fill in the template: your domain, your question, how many sources you read, the tool you used, and what you'll examine. Leave "Findings will show" blank – that comes once you have real results.



Expand Your Introduction into Prose – 100–200 Words

Write three parts – Hook → Context → Research Question – referencing at least two sources. Keep every sentence in your own voice; copied phrasing means you haven't fully understood it.



Voice integrity rule: Don't copy phrases from sources. If you can't say it in your own words, you don't understand it yet - go re-read.

Bring to Session 23

Bring **all** of this to next session:

Source List

With one-sentence summaries for each source

Abstract Sketch

~80 words placeholder

Introduction Draft

100–200 words, in prose form

Final Hands-On Plan

With any adjustments from today

① Your mentor will read your Introduction draft **before** the session - expect detailed feedback.

Next Time - Session 23: Research & Build

The real investigation happens. Together we'll:

Deep-Read

All 5–8 sources in NotebookLM

Run Hands-On Exploration

Test your AI tool!

Draft Methodology

Write how you investigated

Start Results

Begin capturing your findings

- 👉 By the end of Session 23, you'll have **4 sections** of your paper drafted. ✨ Refine your Related Work with the deeper reading too.